A1_PythonIntro_F25

A review of BIT 1401….pretend we learned python last term. 😊

Assignment deliverable:

You will start writing python….normally this would be a review. Your program will:

1. Get input from the user (using input) and outputting messages (using print)
2. Write simple function to calculate the average grade and whether a student “passed the course”
3. Ideally test user input to ensure the data is valid.

This is a one week assignment just to get your feet wet.

Learning Objectives:

By completing the assignment, you will:

- Get experience getting input from the user
- Write functions that have input parameters or return a value.
- Implement for loops and if statements in python (rather than C)
- Perhaps more importantly the format for a duck-typed scripting language takes some time to get used to.
- You will invariably learn how to deal with program errors and debug code so that it will be easier in later assignments.
- ……become more accustomed to the general format of my assignments.

Related Lecture Material:

Conditionals, user input/output, loops and functions.

Grading

All assignments are graded as 25% code running correctly and 75% for the code quality. Points are indicated below.

- Your code should ALWAYS be tested to ensure it works.
- Comments are not only encouraged but you can lose points if your code’s logic is not clear and you don’t explain it
- Do <u>not</u> comment code for the sake of commenting such as the following:
    - my_num = 4 # Set variable my_num to 4

Instructions

1. Create a new **blank** project in PyCharm or the IDE of your choice. We will also do this in the lab and demonstrate this in the lectures.
2. Make a new .py file named A1_<Name>_PyIntro.py
    - The <Name> should be your first initial and last name
    - Write a simple print call to make sure your code runs:

print(“Hello A1!”) Make sure your program works.

3. (20 points) Write a function **get_first_name** that gets a string variable first_name from the user using the input function. The function should return *first_name*.
    - You should keep asking the user for a valid first name until first_name's length is > 0
    - Function first_name.len() returns the number of characters in first_name.
4. (20 points) Write a function **get_last_name** that get the last name from the user and returns that.
    - If the last name has length < 1 output a message indicating there is no last name.
5. In your main code (code not in a function at the bottom of the file) call get_first_name and get_last_name and print the user's full name in a message
6. (20 points) Write a function **get_average** which gets the student's average (using input and converting the string to an int using int()
7. (20 points) Write a function **is_pass_msg(grade)** that takes an integer grade as a parameter and outputs a message whether the student passed the course or not.
    - For example is_passing_msg(49) might output "You failed ITEC 2401. Sorry"
    - Is_passing_msg(82) might output "You passed the course with an 82%. Nice work".
    - The choice of message is up to you.
8. (20 points) A range is a sequence of ordered values from some minimum to some maximum. Thus range(10, 200) is the numbers from 10 to 199. Write a function **print_range(min, max)** that prints out the numbers between min and max (excluding max)

Notes:

- If you are confused, please ask clarification questions on the forum (don't give your code or give away answers). This requires you to start working early.
- Questions and instructions can seem obvious to me but make no sense to students and everyone is confused until someone asks. Please ask.
- Make sure you test that all your functions work. You don't want to assume your code works but it doesn't and the TA gives you no points for the code working.

Submitting:

General Instructions

- When complete, you should make sure your .py file is named A1F25_ <Name>_PyIntro.py
    - <Name> is your first initial and last name. You will lose 10% if you don't do this. My file would be A1F25_MNiknam_PyIntro.py

Confirm everything still works by running your code again.

- Finally, submit your .py file (only) to the BrightSpace Assignment 1 drop box.
- **Remember, there are no late extensions and late assignments are marked as 0.**
- You can also submit multiple times and we grade the last one so <u>SUBMIT EARLY, SUBMIT OFTEN.</u> Hence there is no reason for a late submission.